SAIF Security & Innovation Fair معرض ومسابقة الأمن والابتكار العالمية

Access and download your own copy here

الشريك الرئيسي elm | الشريك المؤسس تحكم TAHAKOM | الشريك الاستراتيجي وزارة التعليم Ministry of Education | تحالف tahaluf | أكاديمية طويق Tuwaiq Academy | بتنظيم من

QUANTUM LEARNING LAB

Kam

Al-Rowad Private Secondary School — Al-Izdihar District — Riyadh

Track 13 — Quantum Technologies & Advanced Computing · Ages 12–17

Riyadh — Kingdom of Saudi Arabia

“Kam” — an Arabic platform that turns quantum computing into a hands-on experience

From concept to circuit in three steps — no advanced mathematics

Students: Osama Hamad Abdulrahman Al-Dailel · Abdullah Omar Mohammed Al-Musnad · Fares Dawood Abdullah Al-Rajhi

Supervisor: Mr. Mohammed El-Sayed Metwally — Computer Science Teacher

Booth —

ID: _____

INTRODUCTION

Quantum computing presents ideas unlike anything we know: a qubit that is 0 and 1 at the same time, and entanglement that links two qubits even when they are far apart.

The problem

- Global simulators are in English, so students stop at the terminology before understanding it.
- Concepts arrive as abstract equations, so students memorize the rule without seeing its effect.
- No single Arabic experience brings explanation, experiment and verification together.

Our question

How do we turn a concept into a short experiment that a twelve-year-old understands, without advanced mathematics?

Who is it for? Students in grades 7–11 (ages 12–17), and teachers who want a ready first quantum lesson without complex preparation.

METHODOLOGY

We built “Kam” in three stages, each reviewed before moving to the next.

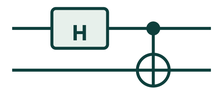
1 Analyse

We reviewed several global simulators and compared content, language and how each one presents the concept.



2 Build

We added a quantum simulator from scratch inside the browser, with a full Arabic interface and no external libraries.



3 Verify

20 automated tests, and 23 circuits compared against the global Quirk simulator.



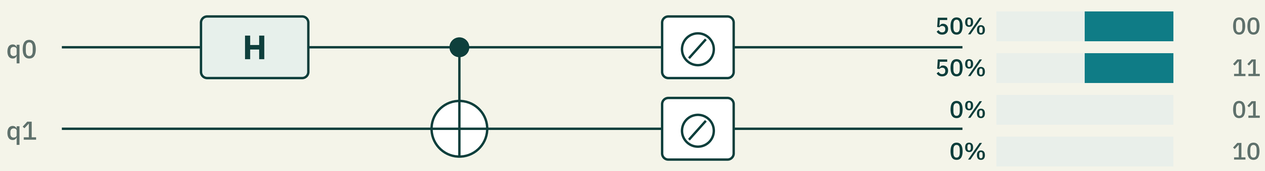
Every page of the platform follows the same path

Understand → Try → Verify

RESULTS

6 core concepts	**8** quantum gates	**6** ready experiments
20 interactive questions	**3** worksheets	**20/20** passing tests

Bell circuit: gate H then CNOT — an even split between 00 and 11 only



What does the result mean? If we repeat the experiment a hundred times, “00” arrives about fifty times and “11” about fifty times, while “01” and “10” never appear — that is entanglement itself.

INNOVATION

A complete Arabic experience: explanation, simulation and verification on a single page.

Works instantly: on a computer or a phone, with no installation or account.

External verification: every circuit is compared against the global Quirk simulator, so nobody is asked to trust us without evidence.

Transparency: zero external libraries, and the code is open to review.

> **The difference in how it is built**
> Students do not read about superposition or memorize its formula… they watch the probabilities move on screen every time they change a gate.

Criterion	“Kam”	Global simulators
Language	Full Arabic, right to left	English or machine translation
Path to understanding	Visual experiment first, then the symbol	Equations and matrices
Verification	Published comparison with Quirk	Internal, not disclosed
Running it	Browser directly, no account	Software that needs installation

CONCLUSION

After the experiment, students explain superposition in their own words and connect every symbol in the circuit to a result they drew themselves.

We shortened the distance between observation and conclusion to three steps: understand, try, verify.

Simplifying science does not mean dropping its precision — it means ordering its path.

One-line takeaway: a single platform, fully in Arabic, opens the door to quantum computing for a twelve-year-old — and its results are measured in numbers, not opinions.

FUTURE WORK & REFERENCES

Scan to visit the platform

meto2004-lang.github.io/saif

Measuring impact: 30 students, a pre-test and a post-test, 20 questions.

Scaling up: an English edition, teacher pathways, and 3-qubit circuits.

References

Nielsen & Chuang, Quantum Computation and Quantum Information, Cambridge Univ. Press, 2010.

Gidney, C., Quirk: Quantum Circuit Simulator, 2016.

Feynman, R., Simulating Physics with Computers, 1982.

Acknowledgments Statement: Special thanks to our teacher Mr. Mohammed El-Sayed Metwally, to the Al-Rowad Private Secondary School family, and to the SAIF 2026 exhibition and competition organizing committee.